

Janadri Yalla Yashwanth

Software Engineer | Backend + AI Systems | Python · FastAPI · PyTorch · Model Deployment

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SUMMARY

Software engineer and 2025 CS (AI & ML) graduate, CGPA 8.6, with a strong foundation in Data Structures & Algorithms, Operating Systems, DBMS, and OOP. At Marut Drones I build backend systems that deploy and serve ML models at scale — FastAPI microservices with Celery/Redis job orchestration, Dockerized GPU/CPU inference, and CI/CD. I write clean, scalable code and I'm eager to build LLM and GenAI systems. Immediate joiner.

TECHNICAL SKILLS

Languages: Python, Java, C++, C, SQL

CS Fundamentals: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, OOP

Backend & Systems: FastAPI, REST APIs, microservices, async/await, Celery, Redis, distributed task queues, Pydantic, SQLAlchemy

Databases: PostgreSQL, SQL, Redis

AI / ML: PyTorch, deep learning, model deployment & inference at scale, OpenCV, YOLO, SAM-2, DeepForest

GenAI / LLM: Prompt engineering (AI-assisted development with Claude Code); learning RAG, embeddings & AI agents

Cloud / DevOps: Docker, GitHub Actions, CI/CD, AWS S3, Git, Linux

Tools: Git & GitHub, Claude Code, VS Code, Jupyter, Docker, GitHub Actions, Postman, Linux

EXPERIENCE

Software Engineer Trainee — Marut Drones

Feb 2026 – Present

- Built and productionized ML detection services (SAM-2, DeepForest, GroundingDINO/LangSAM) behind a FastAPI backend, serving models at scale with async Celery/Redis job orchestration (live progress, cancellation, RBAC).
- Owned model deployment and inference infrastructure — packaged the PyTorch stack into Docker images shipped via GitHub Actions to systemd-managed GPU and CPU workers.
- Engineered data pipelines streaming multi-gigabyte rasters block-by-block from S3 (GDAL/rasterio), with CRS-aware processing, elevation math, and cross-tile deduplication.
- Debugged production inference issues (CUDA/PyTorch mismatches, GPU-versus-CPU serving, failed migrations) and applied clean-code, testing, and version-control practices across the team.

Machine Learning Intern — Rooman Technologies

Jan 2025 – Mar 2025

- Trained and evaluated a machine-learning classification model (TensorFlow) on healthcare data, reaching 98% accuracy.

SELECTED PROJECTS

Custom Object Detection with YOLO

Python, YOLO (Ultralytics), PyTorch, OpenCV, custom datasets

- Trained and validated custom YOLO detectors on self-labeled datasets for number-plate, vehicle, water-body, and broken-fence detection.
- Owned the full pipeline — data collection, annotation, augmentation, train/validation splits, and per-task tuning.

Flight Log Analyzer — Full-Stack Diagnostics Backend

FastAPI, Python, asyncio, React, Docker

- Built a FastAPI service that ingests raw drone logs and returns generated PDF diagnostics, with a from-scratch binary parser decoding 50+ message types and a 50+ check diagnostic engine.
- Containerized the full stack with Docker/Docker Compose for reproducible deployment (~7,200 lines, solo-built).

EDUCATION

Sri Venkateswara College of Engineering, Tirupati

Dec 2021 – May 2025

B.Tech in Computer Science (Artificial Intelligence & Machine Learning) — CGPA: 8.6

Narayana Junior College, Hyderabad

Jun 2019 – Mar 2021

Intermediate (MPC) — 96%

CERTIFICATIONS

- CS50's Introduction to Programming with Python — Harvard University Online

- [Databases: Topics in SQL — Stanford Online](#)